

Options — Exam Reference Card

All 4 Positions: Identify the Graph, Payoff, and When a Company Uses It

The exam gives you a graph. You must: (1) name the position, (2) explain the payoff logic, (3) give a corporate hedging example.

The 4 Positions at a Glance

Position	You...	Max Loss	Max Gain	Pay/Receive premium?
Long Call	Buy a call	Premium paid	Unlimited	Pay
Short Call	Sell a call	Unlimited	Premium received	Receive
Long Put	Buy a put	Premium paid	Strike price – premium	Pay
Short Put	Sell a put	Strike price – premium	Premium received	Receive

POSITION 1 — Long Call

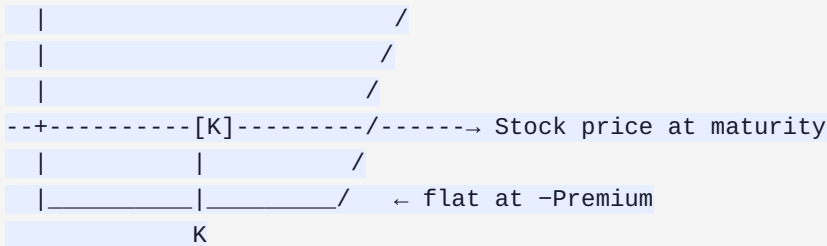
What it is: You buy the RIGHT to purchase an asset at strike price K. You pay a premium upfront.

Payoff at expiry:

If $S > K$: Exercise. Payoff = $S - K$. Profit = $(S - K) - \text{Premium}$
 If $S \leq K$: Don't exercise. Loss = Premium (maximum possible loss)

Graph shape: Flat at $-\text{Premium}$ below K. Then rises linearly above K. Breaks even at $S = K + \text{Premium}$.

Profit



Corporate use — hedges: risk of BUYING something that gets MORE EXPENSIVE

An **importer** (e.g. Belgian company buying goods priced in USD) fears the EUR/USD rate will move against them, making imports costlier. They buy a call on USD — capping their effective exchange rate. If USD stays cheap, they let the option expire and buy at market. If USD spikes, the call compensates.

An **airline** buying jet fuel buys a call on crude oil — capping fuel costs regardless of how high oil goes.

POSITION 2 — Short Call

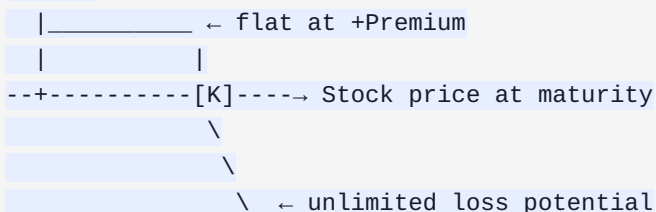
What it is: You SELL a call option. You collect the premium upfront, but take on the obligation to sell the asset at K if the buyer exercises.

Payoff at expiry:

If $S \leq K$: Buyer doesn't exercise. You keep the premium. Profit = Premium
 If $S > K$: Buyer exercises. You must sell at $K < \text{market price}$. Loss = $(S - K) - \text{Premium}$

Graph shape: Flat at $+\text{Premium}$ above K . Then falls linearly below K (mirror image of long call).

Profit



Corporate use — income generation on assets already owned (covered call)

A company that **already holds shares** in another firm sells call options on those shares to earn extra income (the premium). They are willing to sell the shares at K if the price rises that far. Risk: if the stock surges far above K , they miss the upside.

Important: naked short calls (without owning the underlying) are very risky and rarely used for hedging — more of a speculative position.

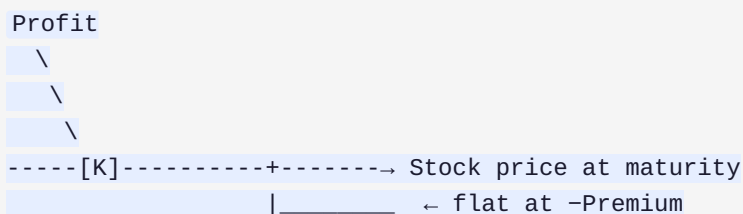
POSITION 3 — Long Put

What it is: You buy the RIGHT to SELL an asset at strike price K . You pay a premium upfront.

Payoff at expiry:

If $S < K$: Exercise. Payoff = $K - S$. Profit = $(K - S) - \text{Premium}$
 If $S \geq K$: Don't exercise. Loss = Premium (maximum possible loss)

Graph shape: Rises linearly as stock falls below K . Flat at $-\text{Premium}$ above K . Mirror image of long call, but on the left side.



Corporate use — hedges: risk of OWNING something that LOSES VALUE (insurance)

A **fund manager** holding a large position in a stock buys put options on it — if the stock crashes, the put pays out, offsetting the loss. This is portfolio insurance.

A **Belgian farmer** growing wheat fears prices will fall before harvest. They buy a put on wheat — guaranteeing a minimum selling price (K), while still benefiting if prices rise (they just don't exercise the put).

An **exporter** receiving USD in 3 months buys a put on USD — locking in a minimum EUR conversion rate. If USD weakens, the put compensates.

POSITION 4 — Short Put

What it is: You SELL a put option. You collect the premium, but take on the obligation to BUY the asset at K if the buyer exercises.

Payoff at expiry:

If $S \geq K$: Buyer doesn't exercise. You keep premium. Profit = Premium
 If $S < K$: Buyer exercises. You must buy at $K >$ market price. Loss = $(K - S) - \text{Premium}$

Graph shape: Flat at +Premium above K. Then falls as stock drops below K.

Profit
 | _____ ← flat at +Premium
 |
 -----[K]-----→ Stock price at maturity
 /
 / ← loss grows as price falls

Corporate use — willingness to acquire at a lower price

A company that **wants to buy shares** in a target (but thinks the current price is too high) sells put options at their desired purchase price K. If the stock falls to K, the put is exercised and they acquire the shares at K — the price they wanted anyway. They also pocket the premium as compensation for waiting.

How to Identify the Graph in the Exam

Step 1 — Look at what happens on the RIGHT side (high stock price): -

Rising line → involves a CALL (benefit from price going UP) - Flat line → involves a PUT (don't benefit from price going up)

Step 2 — Look at the starting level: - Starts BELOW zero (at $-\text{Premium}$) → you BOUGHT the option (Long) - Starts ABOVE zero (at $+\text{Premium}$) → you SOLD the option (Short)

Quick identification table:

Right side	Starting level	Position
Rising (upward slope)	Below zero (−Premium)	Long Call
Falling (downward slope)	Above zero (+Premium)	Short Call
Flat (at +Premium level) with left side rising	Below zero (−Premium)	Long Put
Flat (at +Premium level) with left side falling	Above zero (+Premium)	Short Put

Corporate Hedging — One-Line Rules

You fear...	You use...	Because...
Price of something you BUY goes UP	Long Call	Caps your maximum purchase cost
Price of something you OWN goes DOWN	Long Put	Floors your minimum selling price
You want income on shares you hold	Short Call	Collect premium, agree to sell at K
You want to buy shares cheaper	Short Put	Collect premium, agree to buy at K

Key Terminology the Exam Expects

- **Premium** — the price paid/received for the option contract
- **Strike price (K)** — the pre-agreed exercise price
- **In the money** — exercising would be profitable ($S > K$ for call; $S < K$ for put)
- **Out of the money** — exercising would not be beneficial; option expires worthless
- **Asymmetric payoff** — the option holder has a right, not an obligation → bounded loss, asymmetric upside/downside
- **Intrinsic value** — $\max(S - K, 0)$ for a call; $\max(K - S, 0)$ for a put
- **Hedging vs. speculation** — corporates use options to REDUCE risk exposure, not create it

Study tip: For the exam graph question — identify the shape first (flat vs. sloping on each side), then read the sign of the starting intercept. Those two observations uniquely determine all 4 positions.